



RURO Post-Estimation Analysis

Generated: 2026-05-14 12:54:10

Total Parameters: 48 | **Groups:** joint

Estimation Results Source:

U:\Desktop\Nizam_Hisham\MNL\outputs\estimates\fr\spec\ruro_occ\gamspy\estimation_spec_ruro_occ_M0b2\run_2026-05-14_12-46-04\estimation_results.json

Estimation Configuration

Specification	ruro_occ_M0b2
Specification File	scripts\enhanced\estimation_spec_ruro_occ_M0b2.yaml
Model Family	regular
Proposal Correction	Enabled
Proposal Correction Form	-log(prior)
Opportunity Centering	Enabled

Single Males

Single Females

Males in Couples

Females in Couples

Elapsed Time

Estimation	Post-Estimation	Total	Iterations
4m 54s	20.8s	5m 15s	24



Model Fit Statistics

log_likelihood	-6511.4731
n_observations	4.2530e+05

n_groups	4253.0000
n_parameters	48.0000
AIC	1.3119e+04
BIC	1.3645e+04
ll_null	-1.9586e+04
ll_null_uniform	-1.9586e+04
ll_null_prior_corrected	-2.2322e+04
n_obs_long	4.2530e+05
rho_squared_uniform	0.6675
rho_squared_adj_uniform	0.6651
rho_squared_prior_corrected	0.7083
rho_squared_adj_prior_corrected	0.7061
rho_squared	0.7083
rho_squared_adj	0.7061
AIC_per_obs	0.0308
n_negative_muc_total	0.0000e+00
n_negative_mul_total	0.0000e+00
pct_negative_muc_total	0.0000e+00
pct_negative_mul_total	0.0000e+00
n_bounded_params	48
n_hit_lower_bound	0
n_hit_upper_bound	2



Understanding Bounded Parameters

- **n_bounded_params (48):** Parameters with constraints defined.
- **n_hit_lower_bound (0):** Parameters at lower bound. ✓ Good!

- **n_hit_upper_bound (2):** Parameters at upper bound. ⚠ Check spec

 Identification Diagnostics

Hessian-based diagnostics for parameter identification quality.

Condition Number: 8.52e+09

✗ Severe Identification Problem

The Hessian is severely ill-conditioned ($\kappa \geq 10,000$). This indicates serious identification issues with unreliable parameter estimates.

Eigenvalue Analysis

Minimum Eigenvalue

-1.89e+01

Maximum Eigenvalue

1.36e+10

Negative Eigenvalues

1

✗ NOT a local maximum!

Eigenvalue Summary

p5 / median / p95: 1.71e+00 / 1.89e+02 / 8.89e+05
Smallest 5: -1.89e+01, 1.59e+00, 1.65e+00, 1.90e+00, 5.99e+00
Largest 5: 4.96e+05, 6.18e+05, 9.79e+05, 1.03e+06, 1.36e+10

High Correlations ($|\rho| \geq 0.90$)

Parameter A	Parameter B	Correlation
beta_c_sm	beta_c_sf	-1.069
beta_c_sf	theta_c_singles	-1.054
beta_c_sm	theta_c_singles	-1.042

Parameter A	Parameter B	Correlation
beta_w_pexp	beta_w_pexp2	-0.960
beta_E	beta_E_gsur	-0.950
theta_c_singles	beta_c	-0.904

 Technical Notes

Condition Number (κ): Measures sensitivity of parameter estimates to small changes in data. $\kappa = \lambda_{\max} / \lambda_{\min}$, where λ are eigenvalues of $-H$ (Hessian).

Eigenvalues: All eigenvalues of $-H$ should be positive for a local maximum. Near-zero eigenvalues indicate flat directions (weak identification).

Sign convention: Eigenvalues shown come from the Hessian used to compute SEs. When SEs are computed numerically, this is the Hessian of the negative log-likelihood (H), so negative values indicate H is not positive semidefinite (ill-conditioning or non-optimum).

Standard Errors: Computed as $SE = \sqrt{\text{diag}((-H)^{-1})}$, where H is the Hessian at the optimum. Large SE indicates parameter is not precisely estimated.

 Identification Diagnostics (Estimation Stage)

Source:

U:\Desktop\Nizam_Hisham\MNL\outputs\estimates\fr\spec\ruro_occ\gamspy\estimation_spec_ruro_occ_M0b2\run_2026-05-14_12-46-04\identification_diagnostics.txt

```
specification: ruro_occ_M0b2
n_parameters: 48
bounded_hits_total: 2
bounded_hits_lower: 0
bounded_hits_upper: 2
hessian_condition_number: 8522460122.514409
hessian_negative_eigenvalues: 1
hessian_near_zero_eigenvalues(|eig|<=1e-8): 0
negative_variances_from_varcov: 3

Parameter Stability
```

```
delta_l2: 8.115382e+00
delta_l2_relative: 1.951106e+00
delta_max_abs: 3.382792e+00
delta_mean_abs: 7.250843e-01
```

Bound Hits

```
-----
theta_c: upper (0.0)
beta_l1: upper (2.0)
```

Interpretation hints: many bound hits, negative/near-zero Hessian eigenvalues, or unstable

Model Specification

Utility and opportunity functions for each demographic group, shown in both symbolic and numerical form.

Single Males

Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_1(X) \cdot BC(L, \theta_1)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)

$$\beta_1(X) = \beta_{l0} + \beta_{l,age} \cdot age + \beta_{l,age2} \cdot age^2$$

Utility Function (Numerical)

$$U = 0.5915 \cdot BC(C, 0.0000) + (3.8139 + 0.0082 \cdot age + 0.0020 \cdot age^2) \cdot BC(L, -0.7154)$$

Single Females

Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_1(X) \cdot BC(L, \theta_1)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)

$$\beta_1(X) = \beta_{I0} + \beta_{I,age} \cdot age + \beta_{I,age2} \cdot age2 + \beta_{I,nkids} \cdot nkids$$

Utility Function (Numerical)

$$U = 0.5347 \cdot BC(C, 0.0000) + (4.3828 + 0.0016 \cdot age + 0.0041 \cdot age2 + 0.0633 \cdot nkids) \cdot BC(L, -0.7298)$$

Males in Couples

Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_1(X) \cdot BC(L, \theta_1)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)

$$\beta_1(X) = \beta_{I0} + \beta_{I,age} \cdot age + \beta_{I,age2} \cdot age2$$

Utility Function (Numerical)

$$U = 3.9278 \cdot BC(C, 0.0000) + (0.1254 - 0.0085 \cdot age + 0.0017 \cdot age2) \cdot BC(L, -0.6706)$$

Females in Couples

Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_1(X) \cdot BC(L, \theta_1)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)

$$\beta_1(X) = \beta_{10} + \beta_{1,age} \cdot age + \beta_{1,age2} \cdot age2 + \beta_{1,nkids} \cdot nkids$$

Utility Function (Numerical)

$$U = 3.9278 \cdot BC(C, 0.0000) + (2.8003 - 0.0546 \cdot age + 0.0027 \cdot age2 + 0.1828 \cdot nkids) \cdot BC(L, -0.6534)$$

Model Index

$$\log \text{prior}_{ij} \quad V_{ij} = U_{ij} + O^E_{ij} + O^H_{ij} + O^W_{ij} + O^{Occ}_{ij} + -$$

$$P_{ij} = \exp(V_{ij}) / \sum_{k \in C_i} \exp(V_{ik})$$

U is the utility of consumption and leisure; the O' terms are opportunity-density contributions (employment/hours, wage, occupation); log prior rescales each alternative to the data-generating measure. The terms displayed reflect the opportunity blocks declared in the active YAML specification.



Employment and Hours Opportunity Parameters

Working-gated employment and hours intercepts. Combines `hours_opportunity.shifters` (base employment intercept and PT/FT focal-point indicators) with `market_opportunity.shifters` (working-gated residual market access terms like `gsur`, `education`).

Symbolic form

$$O^E + O^H =$$

```

beta_E * working
+ beta_h_pt1 * working_pt1
+ beta_h_pt2 * working_pt2
+ beta_h_ft * working_ft
+ beta_E_gsur * gsur * working
+ beta_E_educH * educH * working

```

Numerical form (by group)

Joint (All Groups): `-2.8385 * working + -0.5009 * working_pt1 + 0.3707 * working_pt2 + 1.4530 * working_ft + -0.7433 * gsur * working + 0.6129 * educH * working`



Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: $\log(\text{wage}) = \mu_w(X) + \varepsilon, \varepsilon \sim N(0, \sigma^2)$. The mean μ_w is built from `wage_opportunity.mean_shifters`.

Symbolic form

$$\begin{aligned}\mu_w = & \\ & \text{beta_w0} \\ & + \text{beta_w_educL} \cdot \text{educL} \\ & + \text{beta_w_educH} \cdot \text{educH} \\ & + \text{beta_w_pexp} \cdot \text{pexp_years} \\ & + \text{beta_w_pexp2} \cdot \text{pexp_years2} \\ \log(\text{wage}) = \mu_w + \varepsilon, \quad \varepsilon \sim N(0, \text{sigma}^2)\end{aligned}$$

Numerical form (by group)

Joint (All Groups): $\mu_w = 2.0302 + -0.0516 * \text{educL} + 0.3175 * \text{educH} + 0.0181 * \text{pexp_years} + -0.0002 * \text{pexp_years2}$
 $\text{sigma} = 0.4279$



Occupation Opportunity Parameters

Working-gated occupation shifters. Each `applies_to` group has its own coefficient set on the non-reference categories of `loc4`. Reference category: `loc4 = 1`.

Single Males (`applies_to = sm`)

$$\begin{aligned}O_{sm}^{Occ} = & \\ & \text{beta_occ_2_sm} * \text{loc4_2} * \text{working} \\ & + \text{beta_occ_3_sm} * \text{loc4_3} * \text{working} \\ & + \text{beta_occ_4_sm} * \text{loc4_4} * \text{working}\end{aligned}$$

$$\begin{aligned} & -1.5101 * \text{loc4_2} * \text{working} + -2.1649 * \text{loc4_3} * \\ & \text{working} + 0.0240 * \text{loc4_4} * \text{working} \end{aligned}$$

Single Females (applies_to = sf)

$$O^{\text{occ}}_{\text{sf}} =$$

$$\begin{aligned} & \text{beta_occ_2_sf} * \text{loc4_2} * \text{working} \\ & + \text{beta_occ_3_sf} * \text{loc4_3} * \text{working} \\ & + \text{beta_occ_4_sf} * \text{loc4_4} * \text{working} \end{aligned}$$

$$\begin{aligned} & -0.0104 * \text{loc4_2} * \text{working} + -0.5606 * \text{loc4_3} * \\ & \text{working} + 0.7987 * \text{loc4_4} * \text{working} \end{aligned}$$

couples_male (applies_to = cm)

$$O^{\text{occ}}_{\text{cm}} =$$

$$\begin{aligned} & \text{beta_occ_2_cm} * \text{loc4_2} * \text{working} \\ & + \text{beta_occ_3_cm} * \text{loc4_3} * \text{working} \\ & + \text{beta_occ_4_cm} * \text{loc4_4} * \text{working} \end{aligned}$$

$$\begin{aligned} & -1.4788 * \text{loc4_2} * \text{working} + -2.2247 * \text{loc4_3} * \\ & \text{working} + 0.4714 * \text{loc4_4} * \text{working} \end{aligned}$$

couples_female (applies_to = cf)

```
Oocccf =  
  
      beta_occ_2_cf * loc4_2 * working  
+ beta_occ_3_cf * loc4_3 * working  
+ beta_occ_4_cf * loc4_4 * working  
  
      0.1800 * loc4_2 * working + -0.2096 * loc4_3 *  
working + 1.1184 * loc4_4 * working
```

 Curvature-Based Heuristics (Structural Elasticity Approximations)

 Interpretation Note

These are **not** true labor supply elasticities. They are heuristic approximations derived from the curvature parameters (θ) of the Box-Cox utility function. The Hicksian approximation is $(1 - \theta_I)$, which measures the curvature of preferences. For rigorous elasticity estimates, use simulation-based methods that account for the full discrete choice structure and budget constraints.

 Labor Supply Elasticities

Group	Hicksian (compensated)	Marshallian (uncompensated)	Participation (extensive)	Intensive (conditional)	θ_I	θ_c	β_I (at median X)	β_c
Single Males	1.715	1.615	0.515	1.201	-0.715	0.000	3.814	0.592

Group	Hicksian (compensated)	Marshallian (uncompensated)	Participation (extensive)	Intensive (conditional)	θ_l	θ_c	β_l (at median X)	β_c
Single Females	1.730	1.630	0.519	1.211	-0.730	0.000	4.383	0.535
Males in Couples	1.671	1.571	0.501	1.169	-0.671	0.000	0.125	3.928
Females in Couples	1.653	1.553	0.496	1.157	-0.653	0.000	2.800	3.928



Fit Diagnostics

Observed vs Predicted

Group	Obs. Participation	Pred. Participation	Obs. Mean Hours	Pred. Mean Hours
Single Males	93.0%	90.9%	39.3	35.8
Single Females	94.0%	95.2%	36.3	35.1
Males in Couples	97.2%	98.2%	41.6	42.6
Females in Couples	96.5%	98.8%	35.6	38.8



Marginal Utility Diagnostics

Analysis at chosen alternatives. Negative values indicate potential specification issues.

Negative MUC/MUL Summary

MUC Behavior Analysis

For well-behaved utility: $MUC > 0$ ($\beta_c > 0$) and diminishing ($\theta_c < 1$)

Group	N Individuals	% Neg MUC	% Neg MUL	MUC Mean	MUL Mean
Single Males	910	0.00%	0.00%	0.0003	7.1251e-03
Single Females	766	0.00%	0.00%	0.0003	9.5971e-03
Males in Couples	2577	0.00%	0.00%	0.0010	3.6610e-04
Females in Couples	2577	0.00%	0.00%	0.0010	6.4339e-03

Group	β_c	θ_c	MUC Positive?	MUC Diminishing?	Well-Behaved?	MUC at Median C	C where MUC=1	Notes
Single Males	0.5915	0.0000	✓	✓	✓	0.5915	0.5915	
Single Females	0.5347	0.0000	✓	✓	✓	0.5347	0.5347	
Males in Couples	3.9278	0.0000	✓	✓	✓	3.9278	3.9278	
Females in Couples	3.9278	0.0000	✓	✓	✓	3.9278	3.9278	

 Choice Probability Diagnostics

Probability Sum Sanity Check

Max $ \Sigma p - 1 $	0.000000
Mean $ \Sigma p - 1 $	0.000000
% HH off by >0.01	0.00%
% HH off by >0.001	0.00%

P(chosen) Distribution

Min	0.0000
10th percentile	0.0162
25th percentile	0.0676
Median	0.2725
Mean	0.3900
75th percentile	0.7270
90th percentile	0.9219
Max	0.9993

 Parameters at Bounds (within 1e-6)

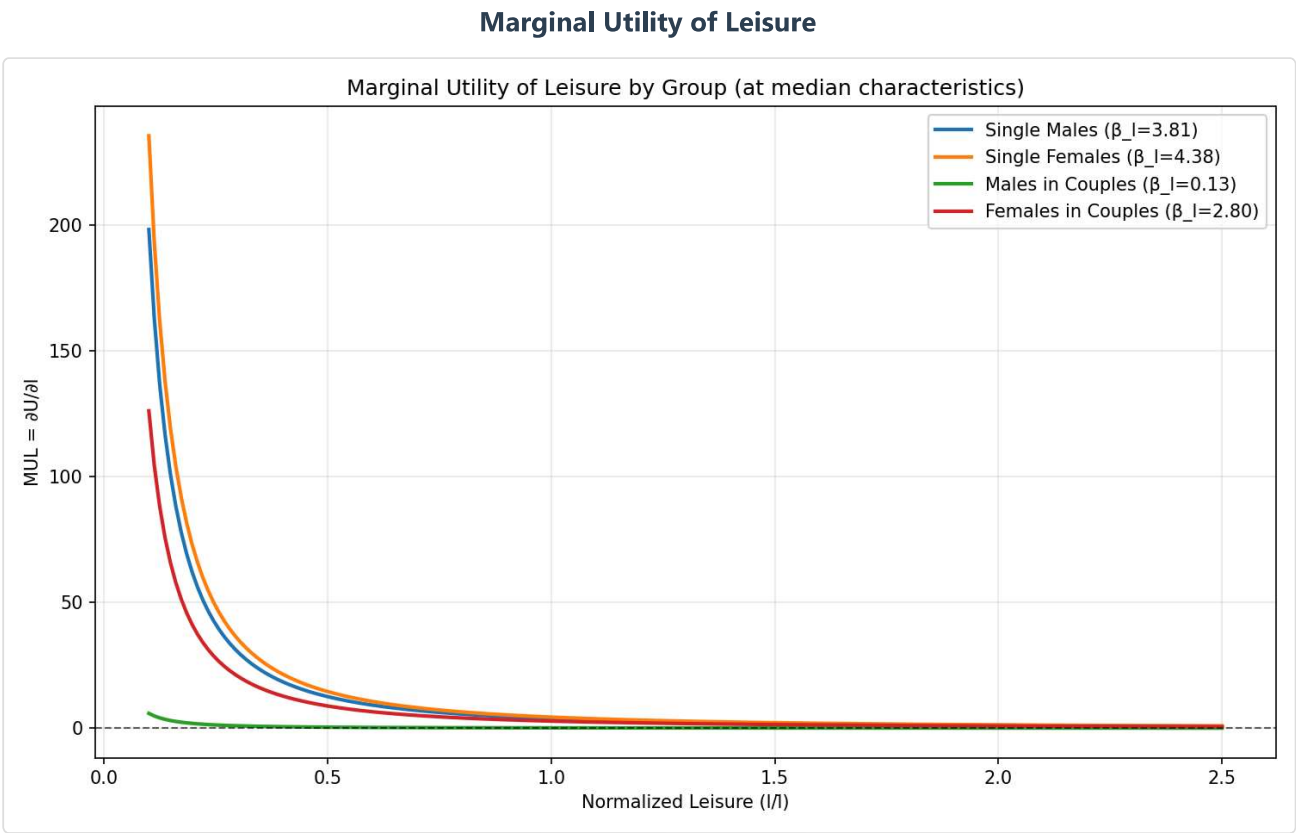
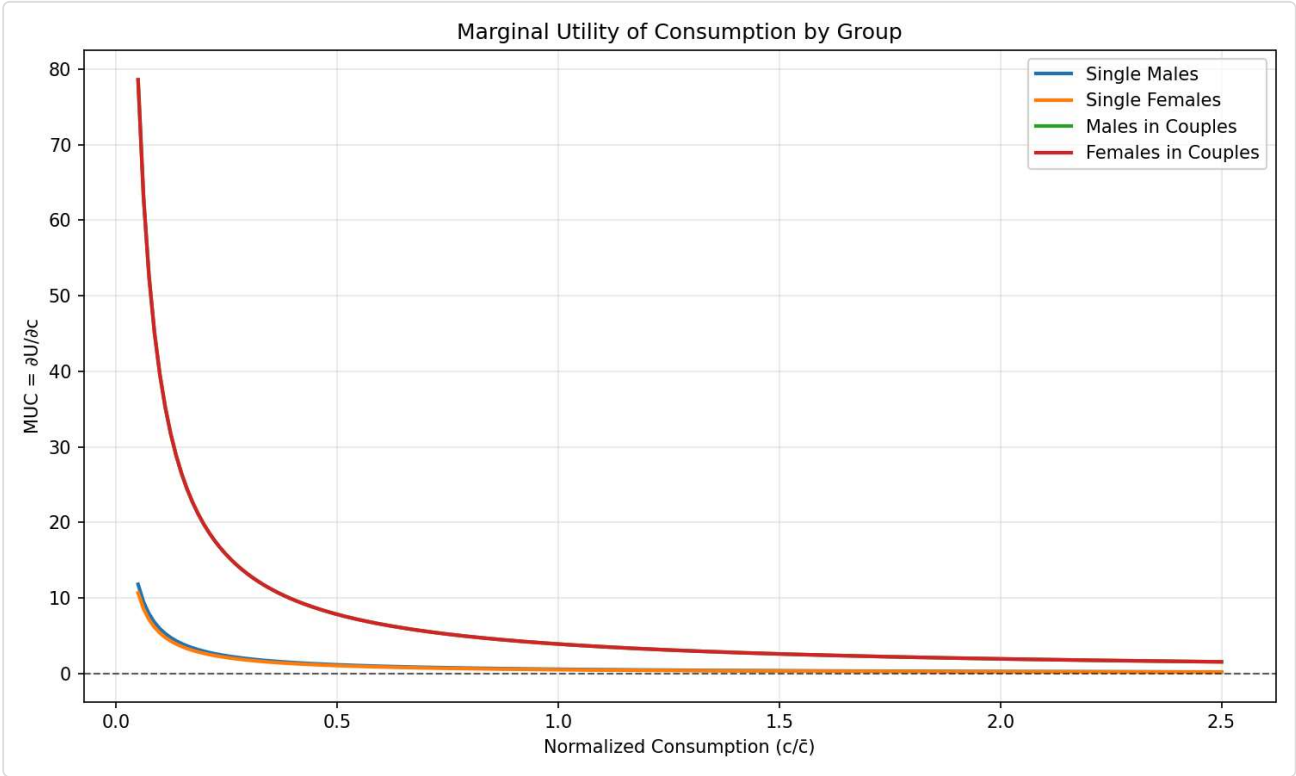
These parameters hit their constraints. Consider checking if bounds are too restrictive.

Parameter	Estimate	Bound	Side
joint.theta_c	0.000000	0.000000	 upper
joint.beta_ll	2.000000	2.000000	 upper

 Utility Contours & Plots

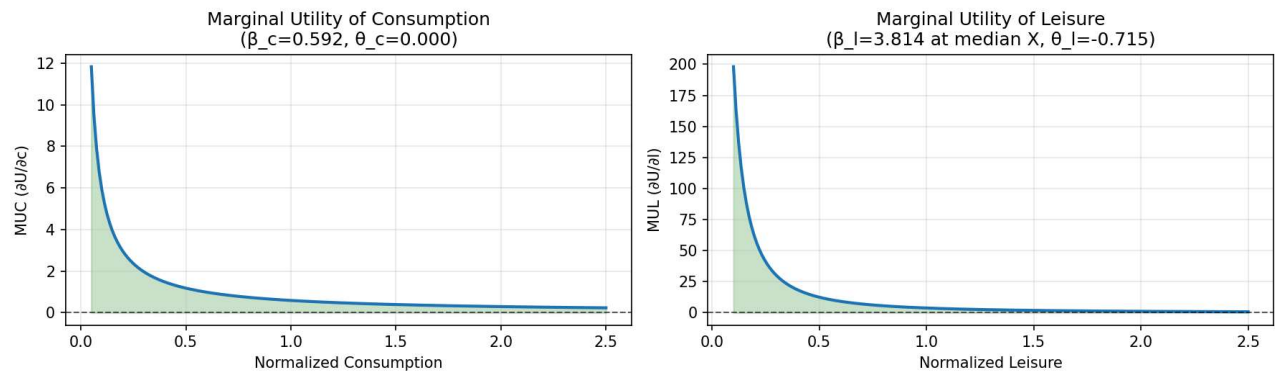
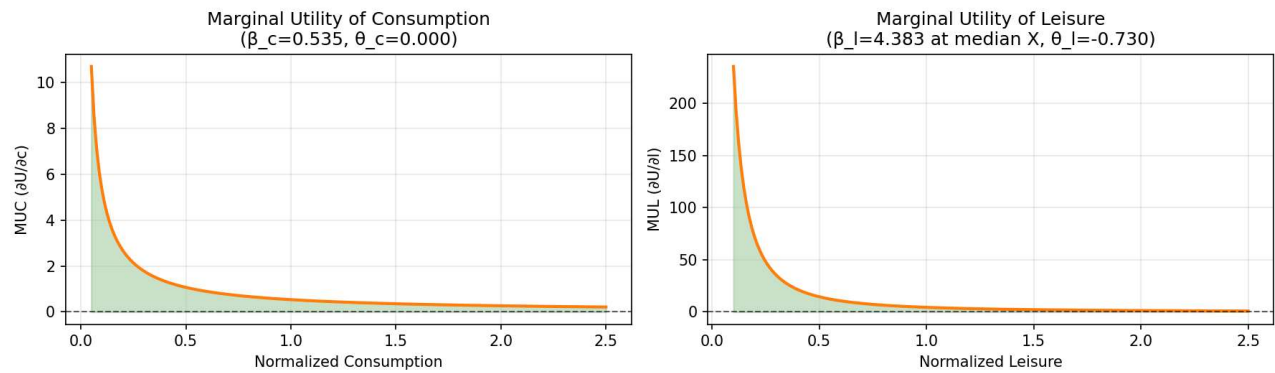
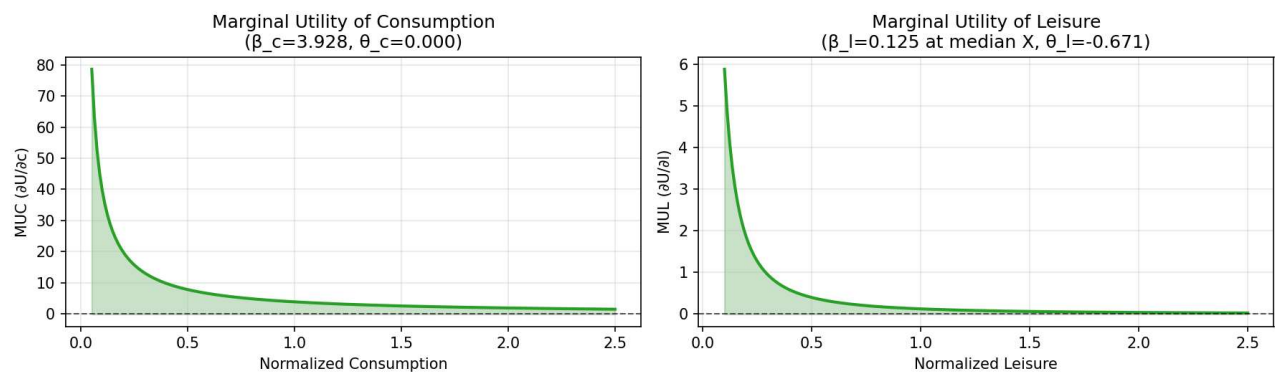
Marginal Utility Comparison

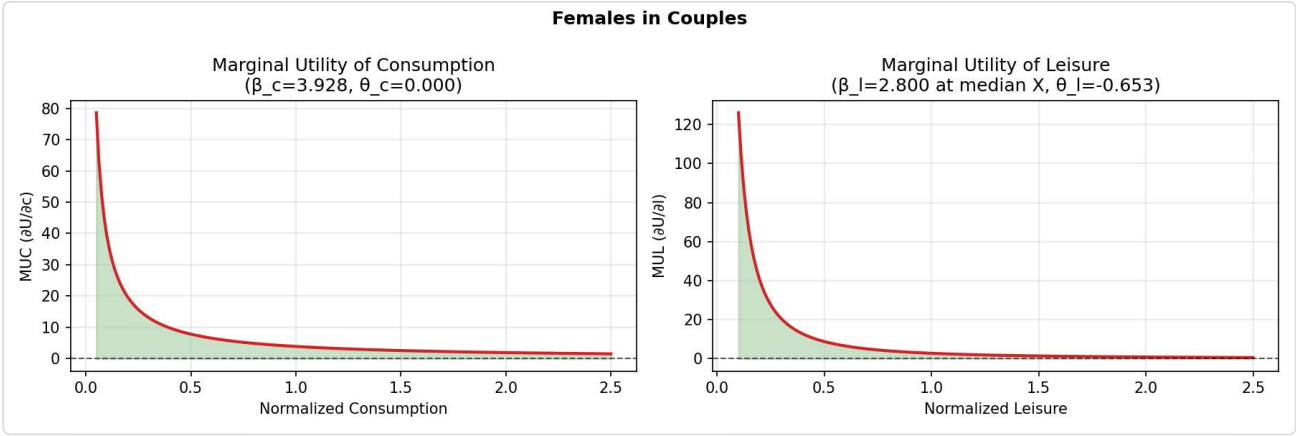
Marginal Utility of Consumption



 Marginal Utility Distributions by Group

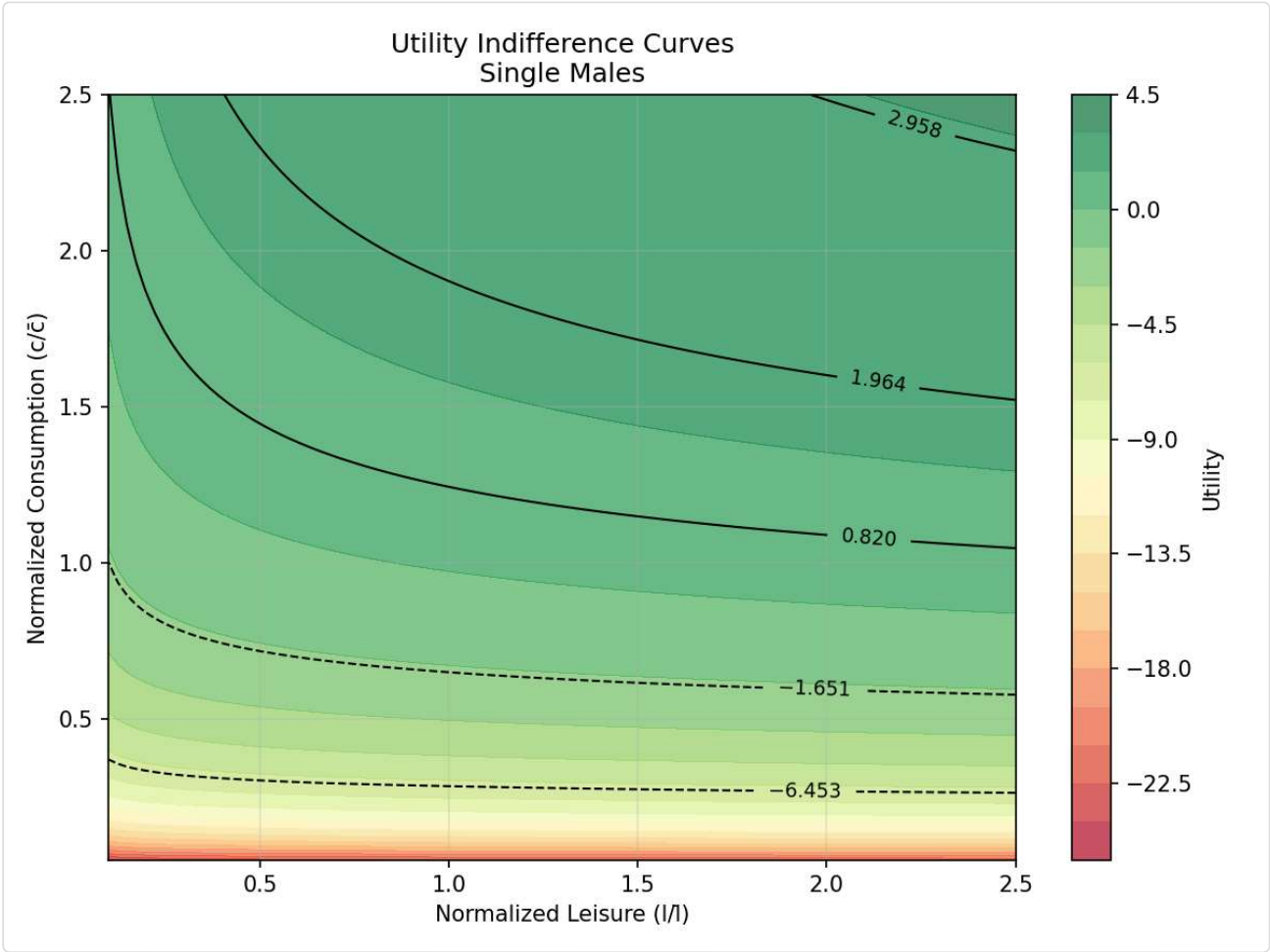
Single Males - MUC & MUL Distributions

Single Males**Single Females - MUC & MUL Distributions****Single Females****Males in Couples - MUC & MUL Distributions****Males in Couples****Females in Couples - MUC & MUL Distributions**

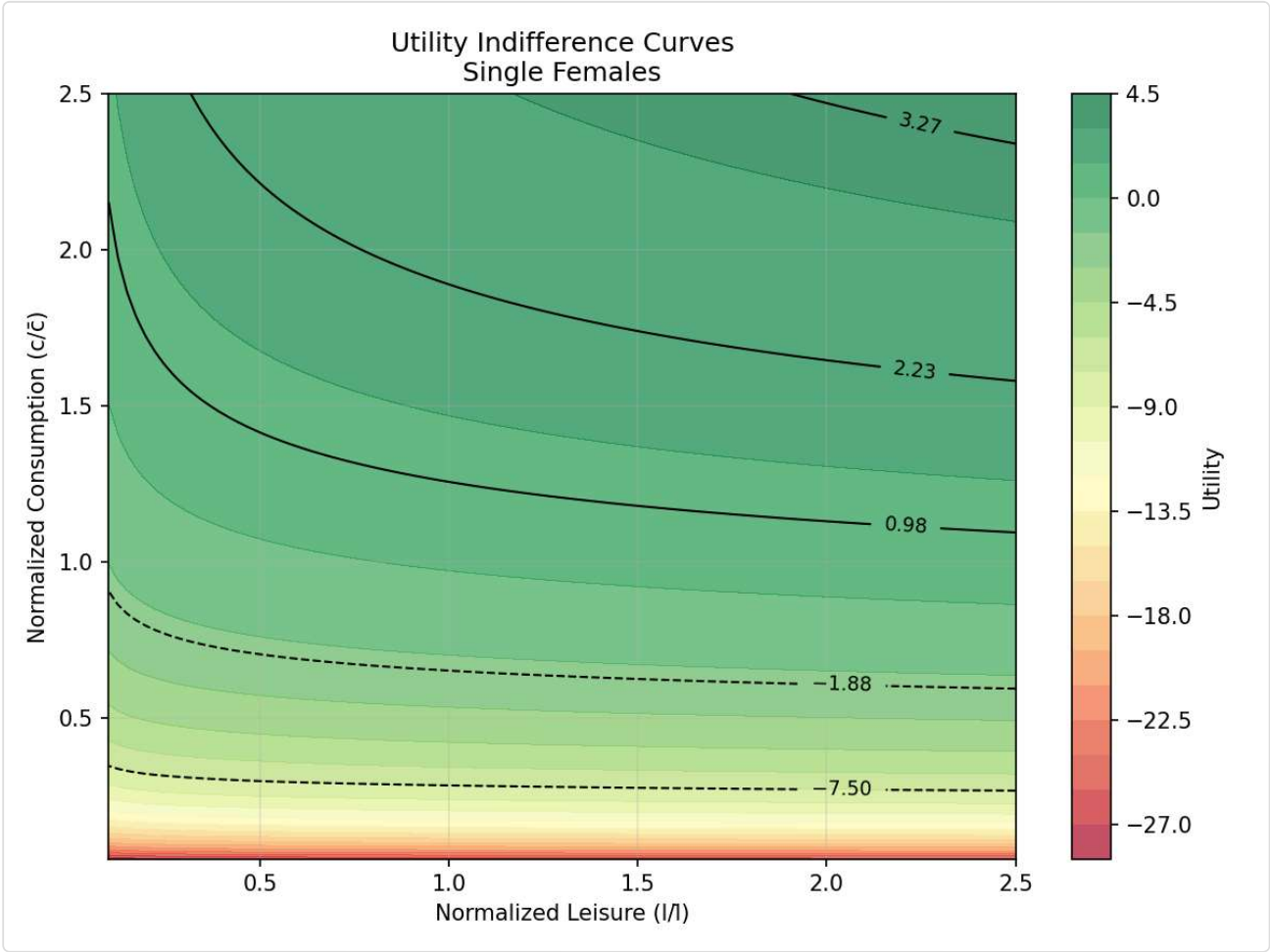


Utility Indifference Curves by Group

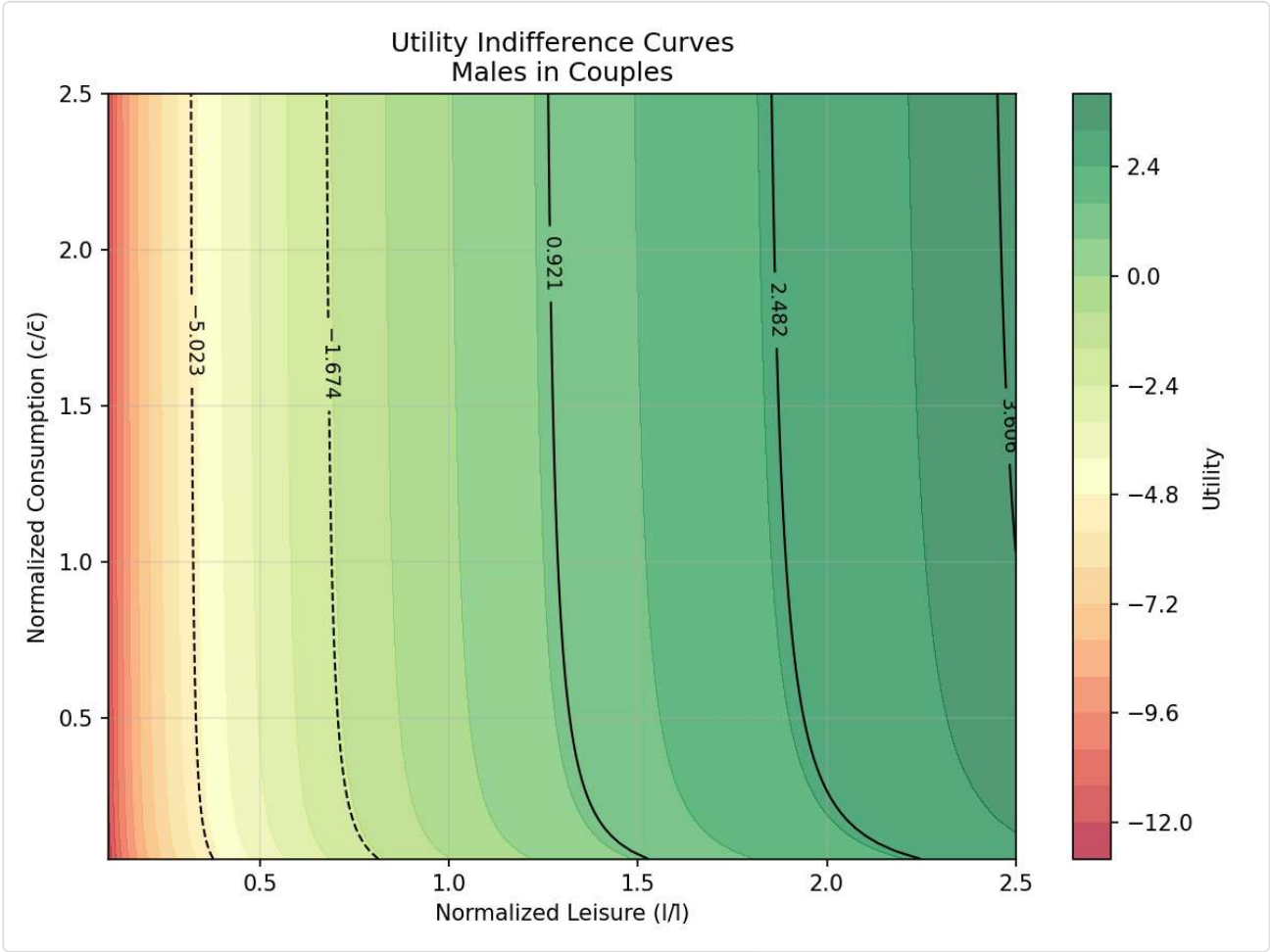
Single Males



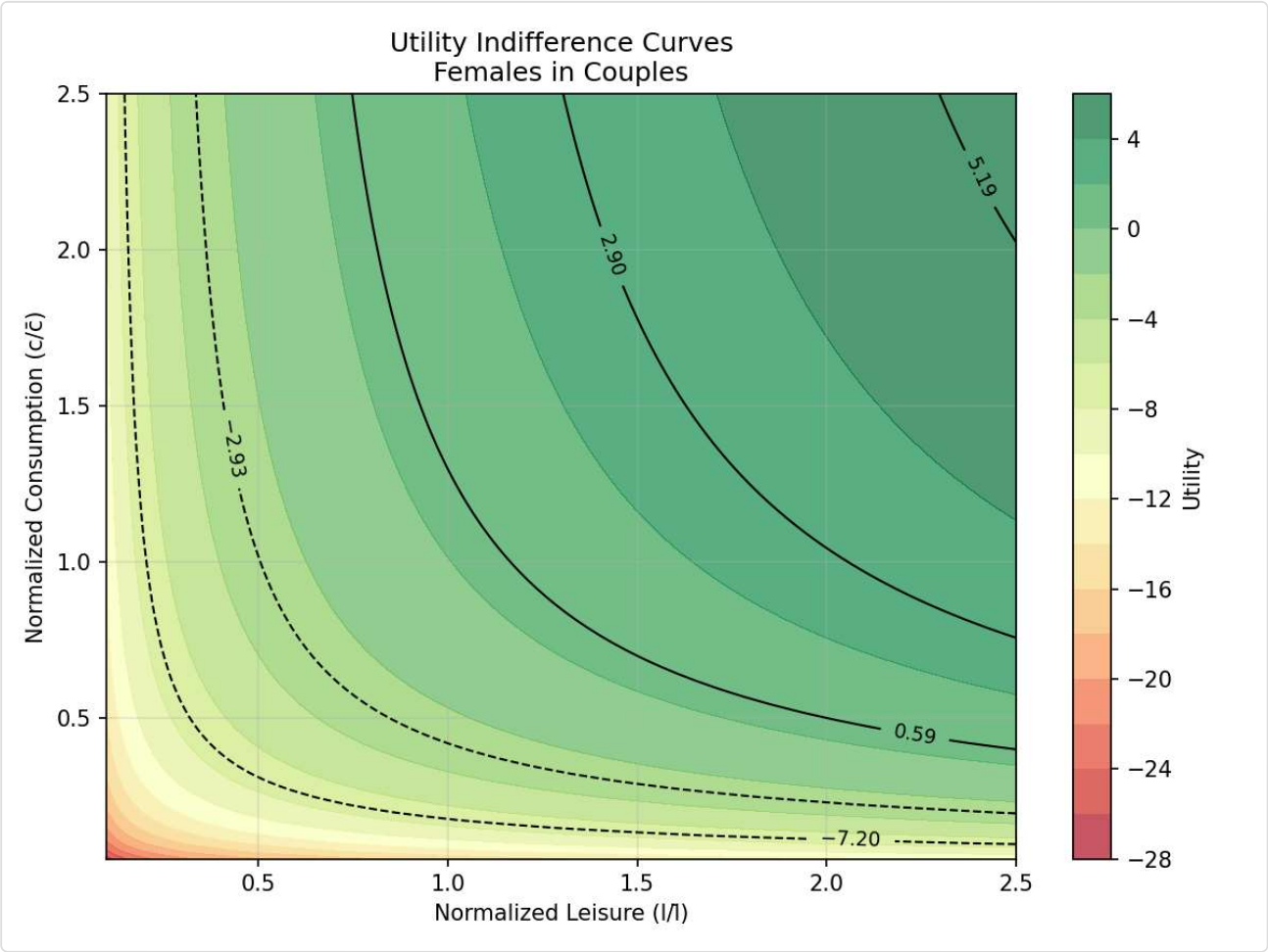
Single Females



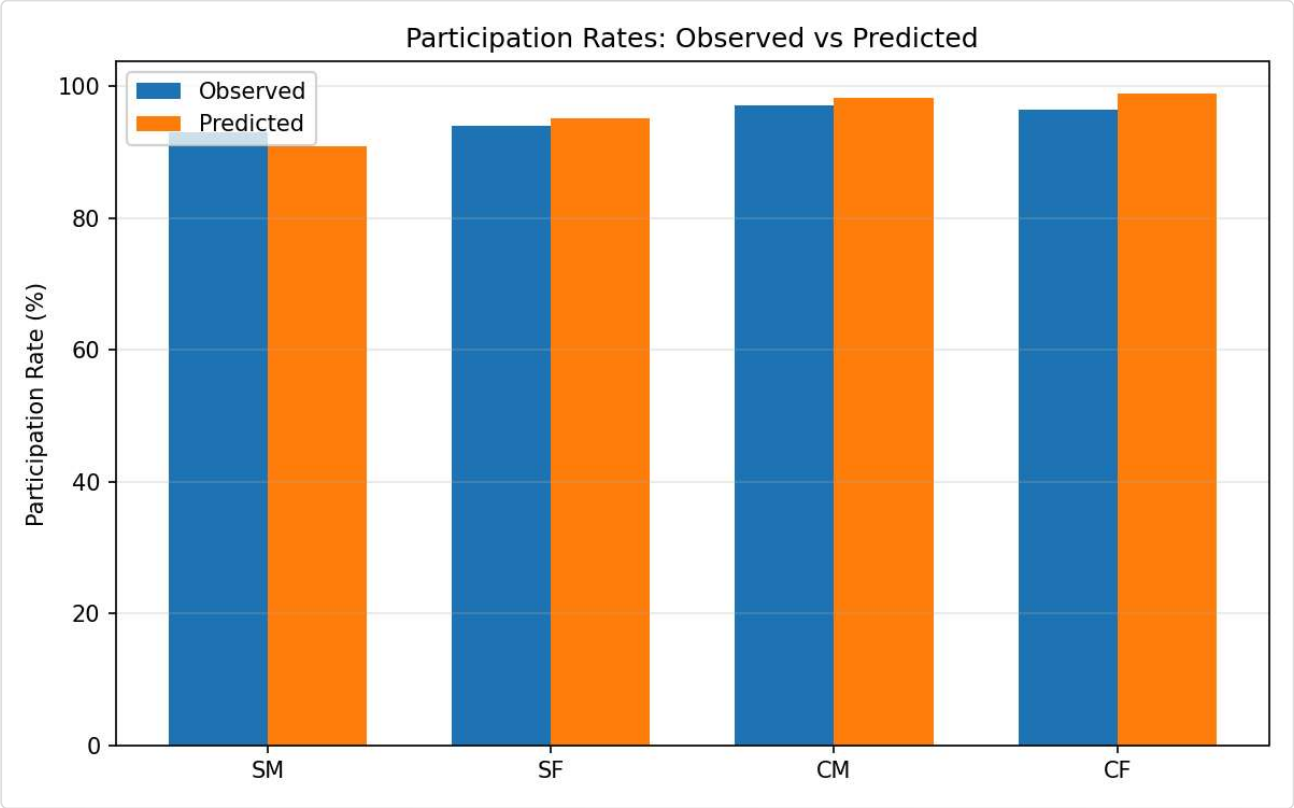
Males in Couples



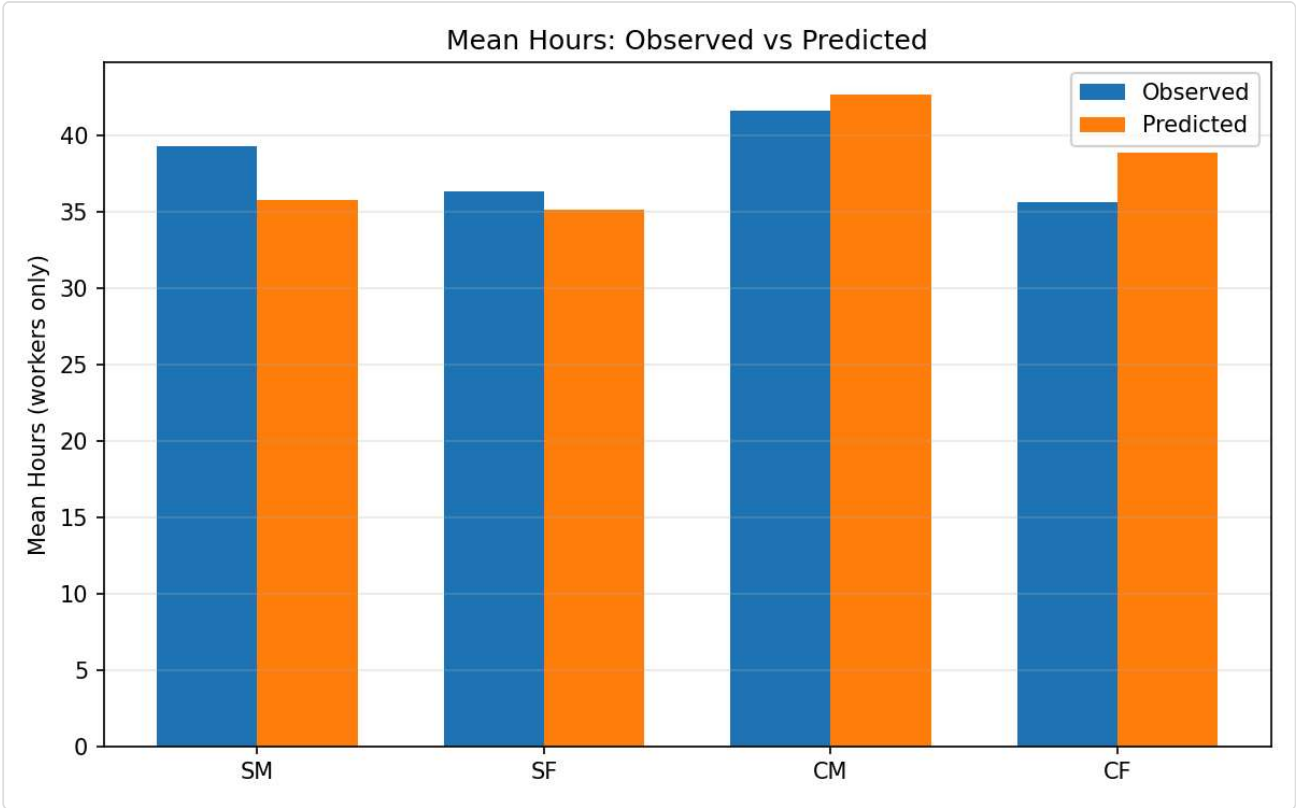
Females in Couples



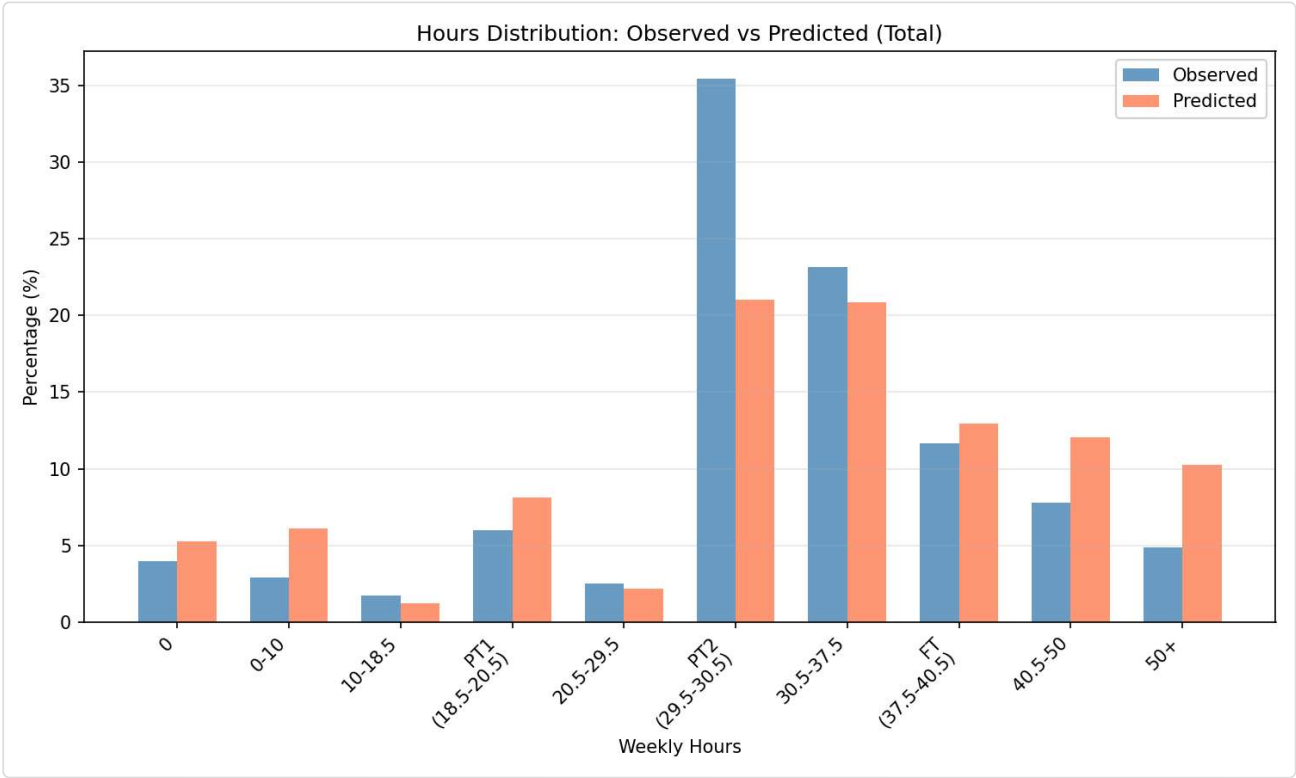
Participation

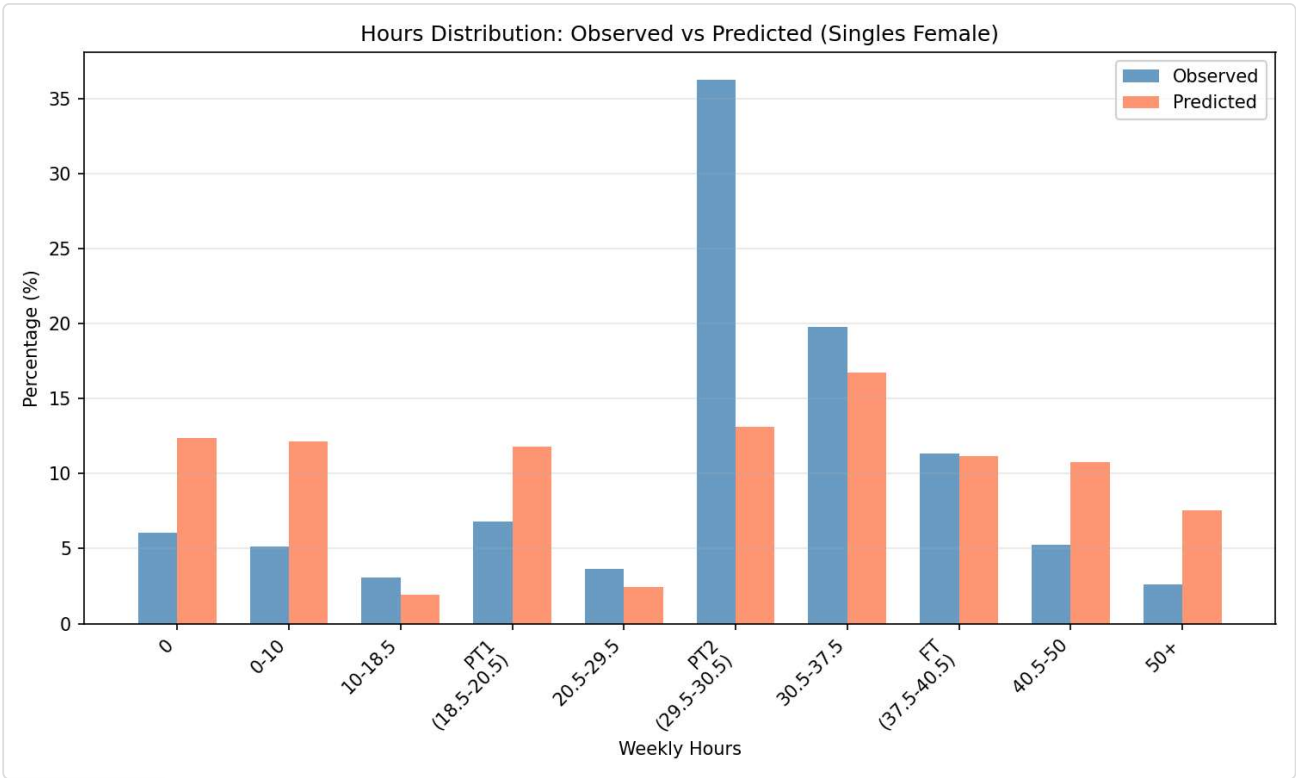
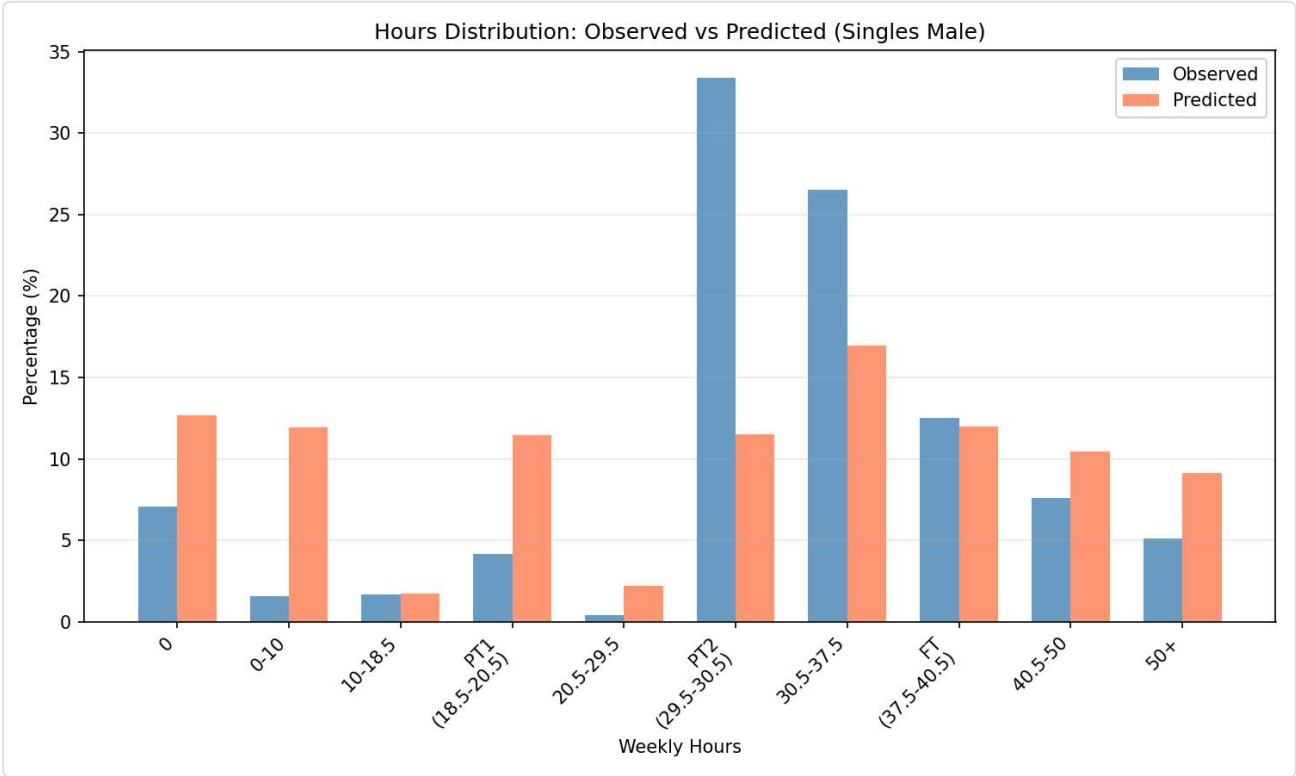


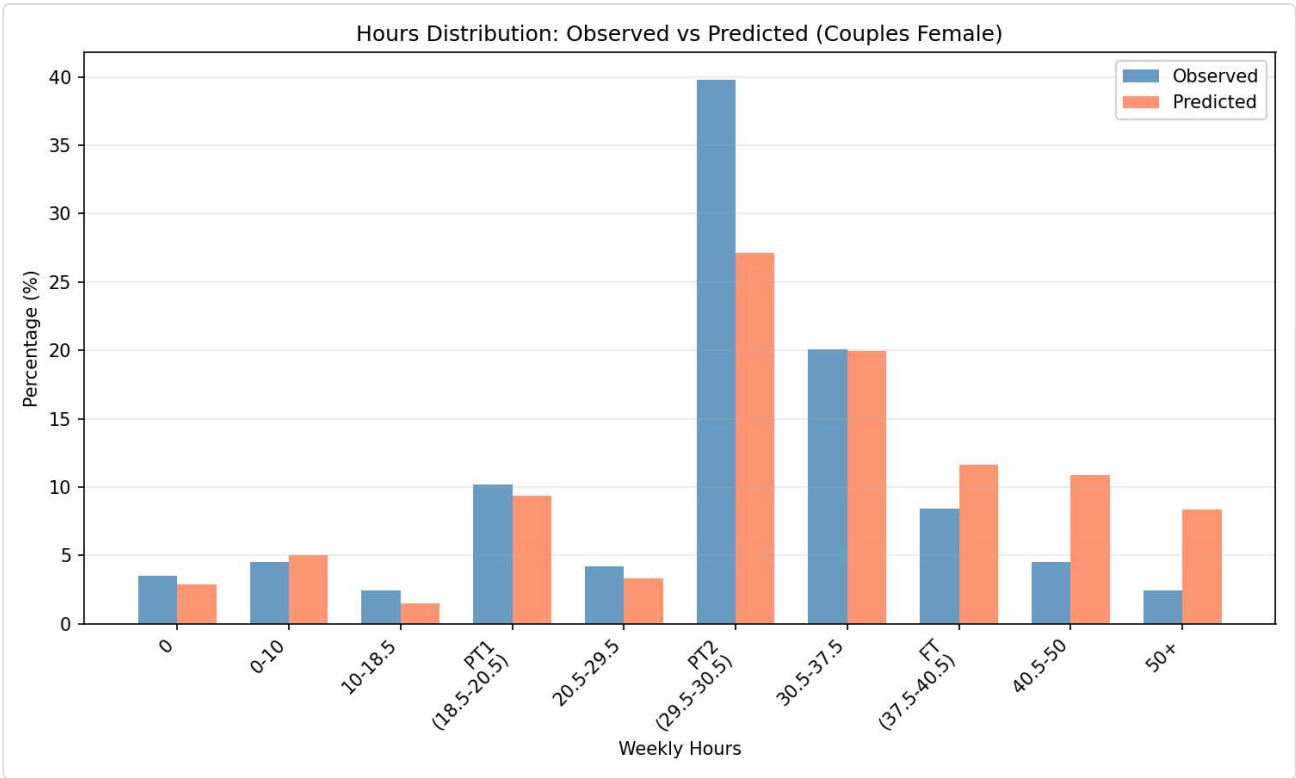
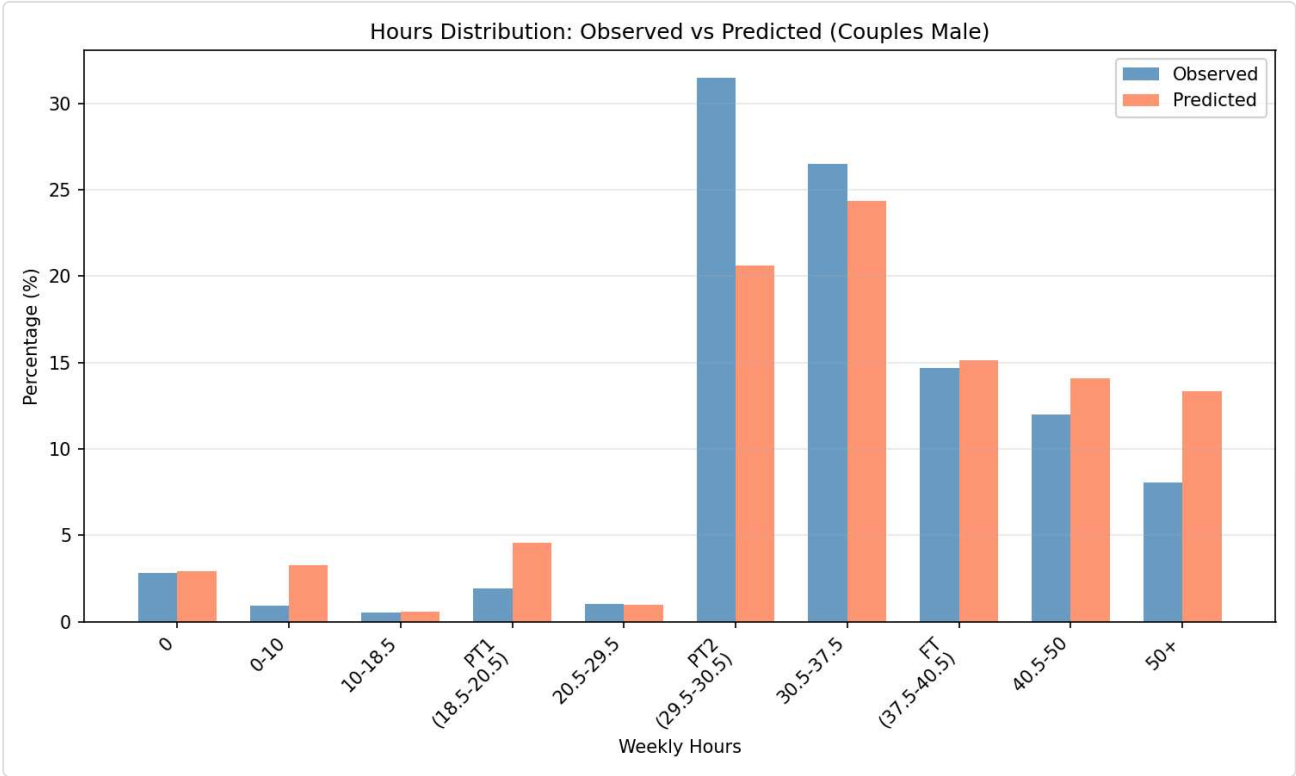
Mean Hours



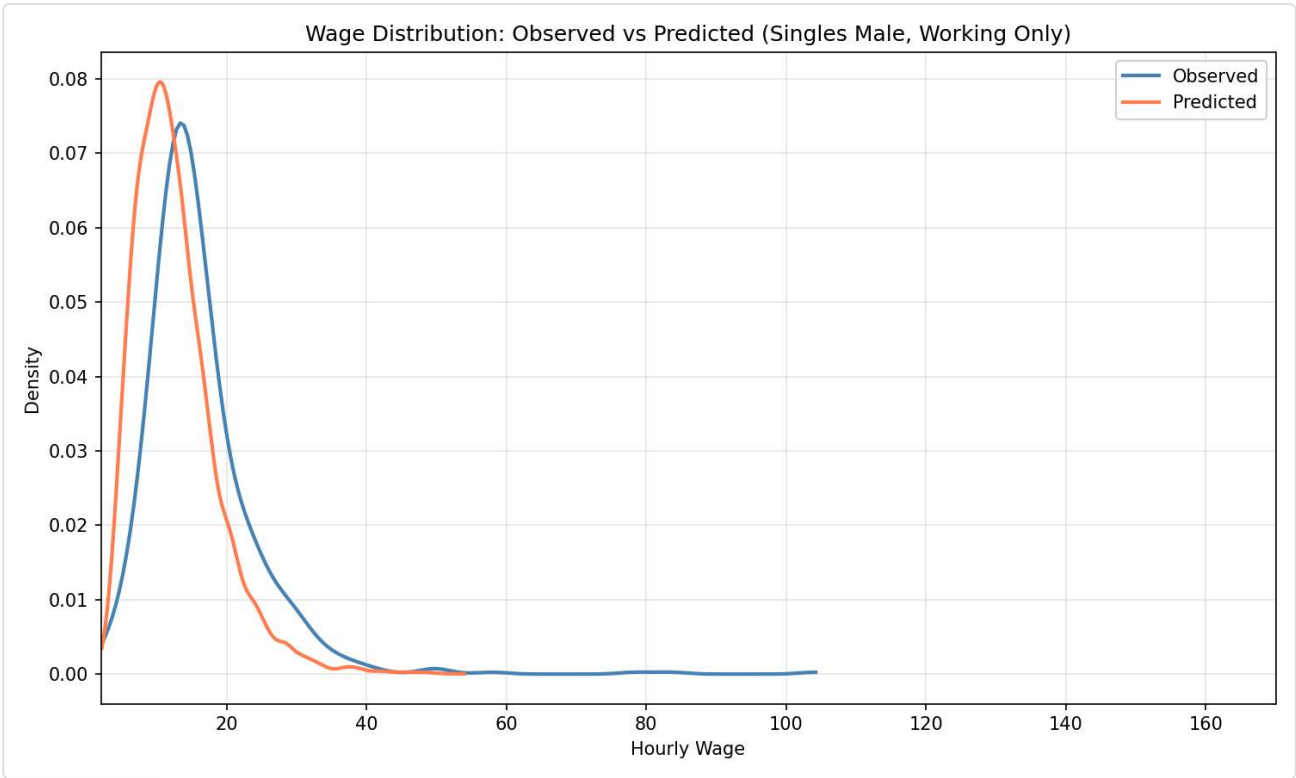
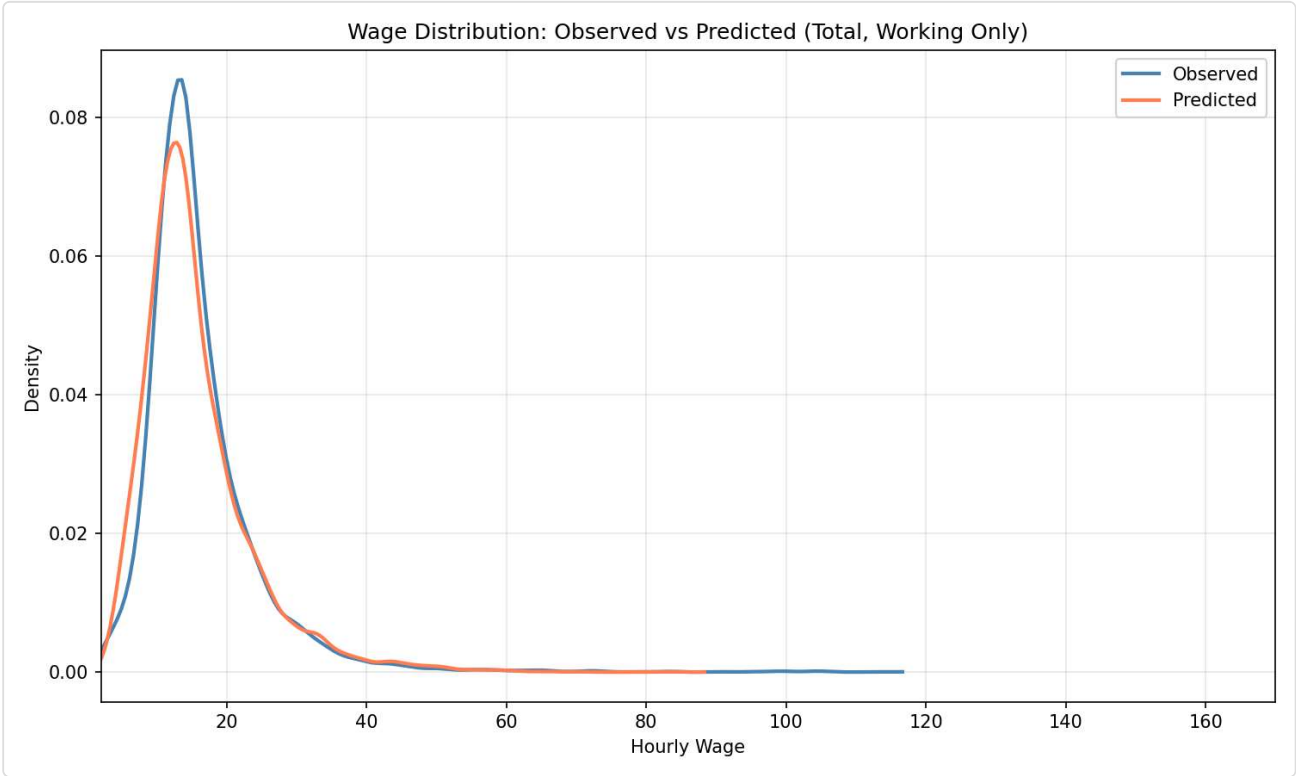
?? Hours Distribution: Observed vs Predicted

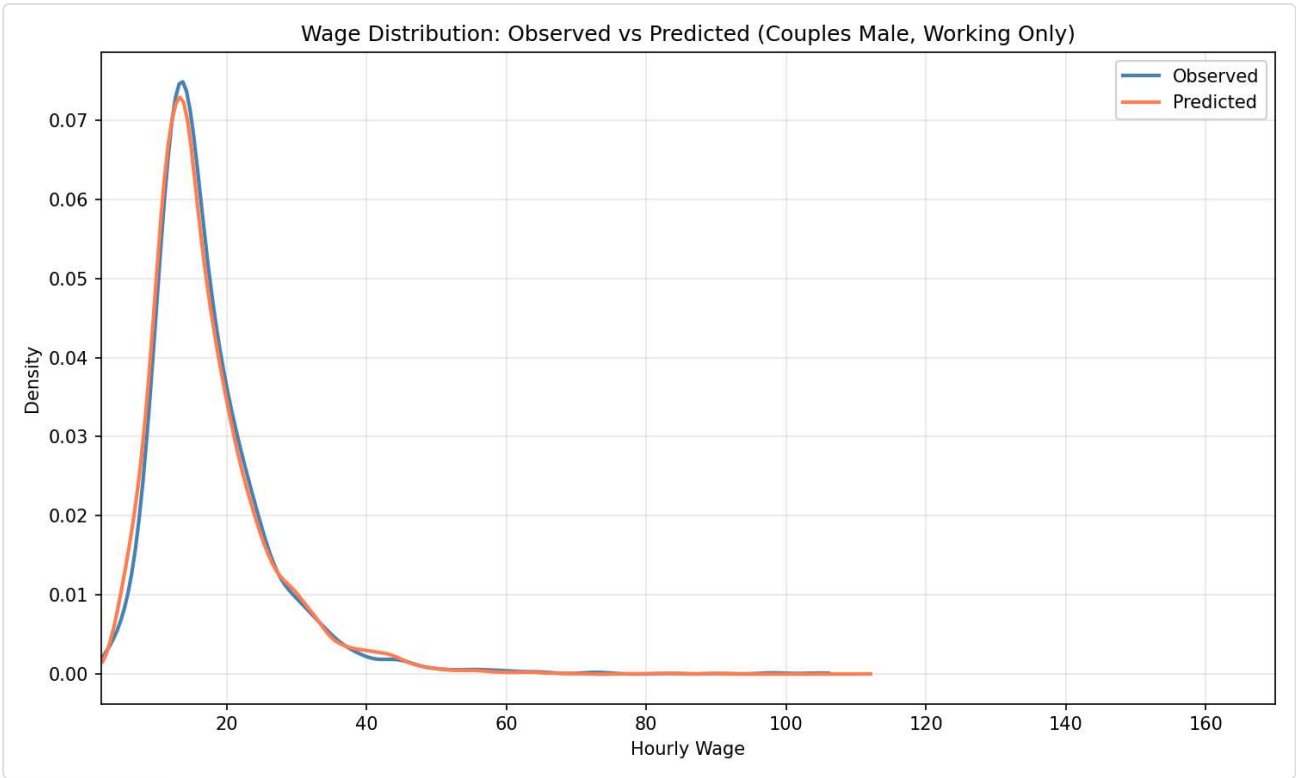
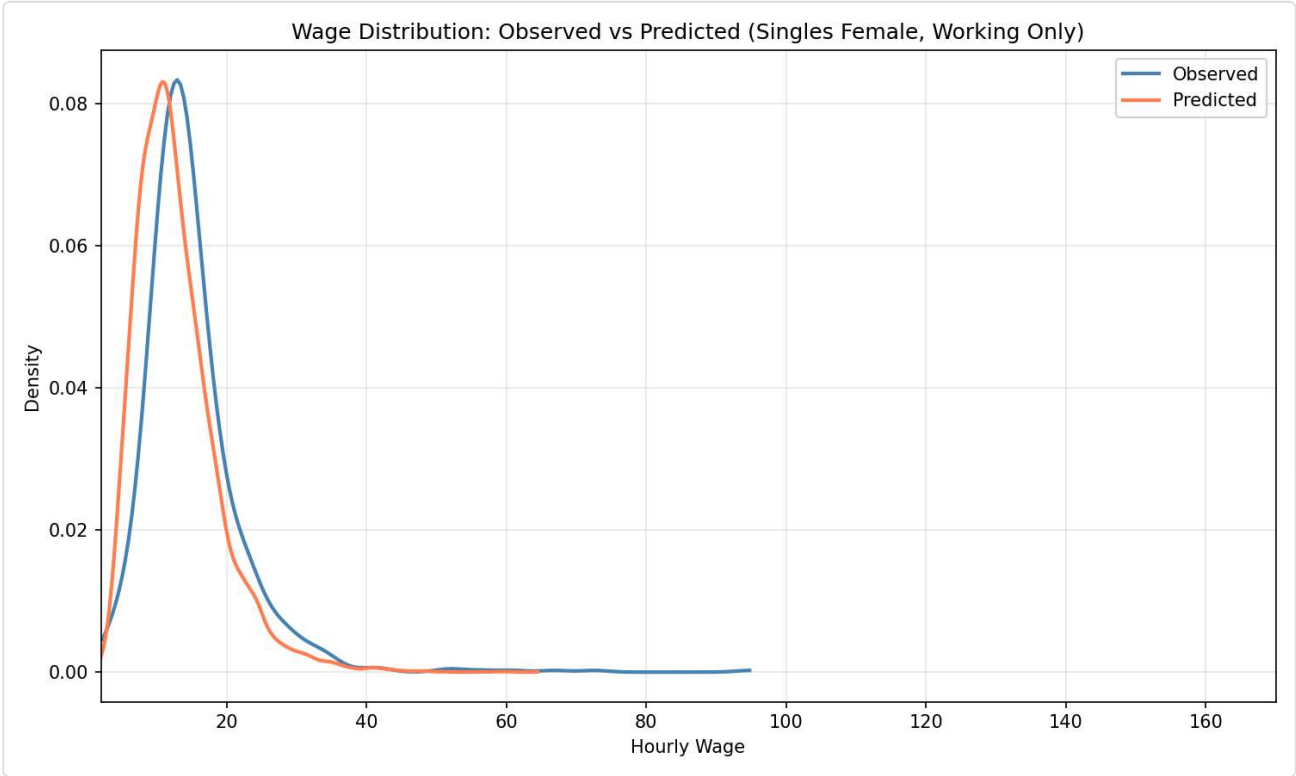


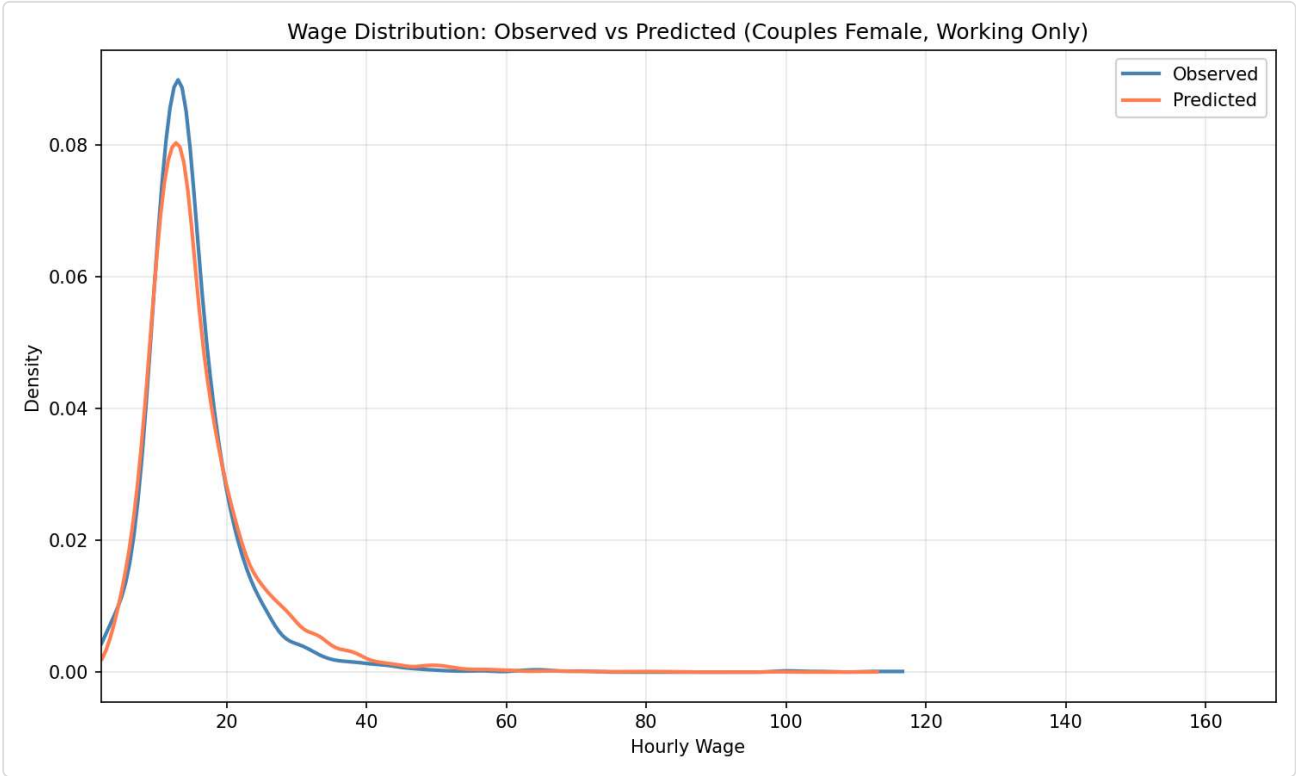




?? Wage Distribution: Observed vs Predicted (Working Only)







Group-Specific Parameters

Single Males

beta_E	-2.8385
beta_E_educH	0.6129
beta_E_gsur	-0.7433
beta_c	0.5915
beta_h_ft	1.4530
beta_h_pt1	-0.5009
beta_h_pt2	0.3707
beta_l0	3.8139
beta_l_age	0.0082
beta_l_age2	0.0020
beta_ll	2.0000

Single Females

beta_E	-2.8385
beta_E_educH	0.6129
beta_E_gsur	-0.7433
beta_c	0.5347
beta_h_ft	1.4530
beta_h_pt1	-0.5009
beta_h_pt2	0.3707
beta_l0	4.3828
beta_l_age	0.0016
beta_l_age2	0.0041
beta_l_nkids	0.0633

beta_occ_2	-1.5101
beta_occ_2_cf	0.1800
beta_occ_2_cm	-1.4788
beta_occ_3	-2.1649
beta_occ_3_cf	-0.2096
beta_occ_3_cm	-2.2247
beta_occ_4	0.0240
beta_occ_4_cf	1.1184
beta_occ_4_cm	0.4714
beta_w0	2.0302
beta_w_educH	0.3175
beta_w_educL	-0.0516
beta_w_pexp	0.0181
beta_w_pexp2	-0.0002
sigma	0.4279
theta_c	0.0000
theta_c_singles	-0.9708
theta_l	-0.7154

beta_ll	2.0000
beta_occ_2	-0.0104
beta_occ_2_cf	0.1800
beta_occ_2_cm	-1.4788
beta_occ_3	-0.5606
beta_occ_3_cf	-0.2096
beta_occ_3_cm	-2.2247
beta_occ_4	0.7987
beta_occ_4_cf	1.1184
beta_occ_4_cm	0.4714
beta_w0	2.0302
beta_w_educH	0.3175
beta_w_educL	-0.0516
beta_w_pexp	0.0181
beta_w_pexp2	-0.0002
sigma	0.4279
theta_c	0.0000
theta_c_singles	-0.9708
theta_l	-0.7298

Males in Couples

beta_E	-2.8385
beta_E_educH	0.6129
beta_E_gsur	-0.7433
beta_c	3.9278
beta_h_ft	1.4530
beta_h_pt1	-0.5009
beta_h_pt2	0.3707
beta_l0	0.1254
beta_l_age	-0.0085
beta_l_age2	0.0017

Females in Couples

beta_E	-2.8385
beta_E_educH	0.6129
beta_E_gsur	-0.7433
beta_c	3.9278
beta_h_ft	1.4530
beta_h_pt1	-0.5009
beta_h_pt2	0.3707
beta_l0	2.8003
beta_l_age	-0.0546
beta_l_age2	0.0027

beta_ll	2.0000
beta_occ_2_cf	0.1800
beta_occ_2_cm	-1.4788
beta_occ_3_cf	-0.2096
beta_occ_3_cm	-2.2247
beta_occ_4_cf	1.1184
beta_occ_4_cm	0.4714
beta_w0	2.0302
beta_w_educH	0.3175
beta_w_educL	-0.0516
beta_w_pexp	0.0181
beta_w_pexp2	-0.0002
sigma	0.4279
theta_c	0.0000
theta_c_singles	-0.9708
theta_l	-0.6706

beta_l_nkids	0.1828
beta_ll	2.0000
beta_occ_2_cf	0.1800
beta_occ_2_cm	-1.4788
beta_occ_3_cf	-0.2096
beta_occ_3_cm	-2.2247
beta_occ_4_cf	1.1184
beta_occ_4_cm	0.4714
beta_w0	2.0302
beta_w_educH	0.3175
beta_w_educL	-0.0516
beta_w_pexp	0.0181
beta_w_pexp2	-0.0002
sigma	0.4279
theta_c	0.0000
theta_c_singles	-0.9708
theta_l	-0.6534

 Parameter Estimates by Category

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

- ☐ Bounded (has constraints)
- ☐  Hit bound
- ☐ Warning: degenerate SE (near zero)
- ☐ $p \in [0.05, 0.1)$
- ☐ $p \in [0.1, 0.25)$
- ☐ $p \geq 0.25$

 Preference Parameters (Utility Function)

Parameters determining the utility from consumption (β_c, θ_c) and leisure (β_l^*, θ_l) by demographic group.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_l0_sm	3.8139	0.7137	5.34	0.0000 ***	0.0500	50.0000	1.0000
joint.beta_l_age_sm	0.0082	0.0248	0.33	0.7410	-5.0000	5.0000	0.0000
joint.beta_l_age2_sm	0.0020	0.0021	0.97	0.3306	-1.0000	1.0000	0.0000
joint.beta_c_sm	0.5915	N/A	N/A	N/A	0.0500	50.0000	1.0000
joint.theta_l_sm	-0.7154	0.1556	-4.60	0.0000 ***	-8.0000	0.9500	-1.0000
joint.beta_l0_sf	4.3828	0.7726	5.67	0.0000 ***	0.0500	50.0000	1.0000
joint.beta_l_age_sf	0.0016	0.0272	0.06	0.9517	-5.0000	5.0000	0.0000
joint.beta_l_age2_sf	0.0041	0.0026	1.61	0.1065	-1.0000	1.0000	0.0000



Employment and Hours Opportunity Parameters

Working-gated employment intercept and hours focal-point indicators, plus residual market-access shifters (*gsur*, *education*) declared in *market_opportunity.shifters*.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_E	-2.8385	0.2980	-9.53	0.0000 ***	-25.0000	25.0000	0.0000
joint.beta_h_pt1	-0.5009	0.1078	-4.64	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_h_pt2	0.3707	0.1111	3.34	0.0009 ***	-10.0000	10.0000	0.0000
joint.beta_h_ft	1.4530	0.0501	29.03	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_E_gsur	-0.7433	0.2197	-3.38	0.0007 ***	-10.0000	10.0000	0.0000
joint.beta_E_educH	0.6129	0.2350	2.61	0.0091 **	-10.0000	10.0000	0.0000



Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: $\mu_w(X)$ intercept, education, experience, and residual standard deviation σ .

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_w0	2.0302	0.0254	79.80	0.0000 ***	-10.0000	20.0000	2.0000
joint.beta_w_educL	-0.0516	0.0212	-2.43	0.0151 *	-5.0000	5.0000	-0.1000
joint.beta_w_educH	0.3175	0.0150	21.12	0.0000 ***	-5.0000	5.0000	0.2000
joint.beta_w_pexp	0.0181	0.0022	8.09	0.0000 ***	-1.0000	1.0000	0.0200
joint.beta_w_pexp2	-0.0002	0.0000	-4.40	0.0000 ***	-0.1000	0.1000	-0.0003

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.sigma	0.4279	0.0041	105.18	0.0000 ***	0.1000	20.0000	0.5000

 Occupation Opportunity Parameters

Working-gated occupation shifters declared in `occupation_opportunity.shifters`; routed by `applies_to` group (`sm/sf/cm/cf`). Reference category is the YAML's `occupation_opportunity.reference`.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_occ_2_sm	-1.5101	0.1420	-10.63	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_3_sm	-2.1649	0.1842	-11.75	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_sm	0.0240	0.0860	0.28	0.7804	-10.0000	10.0000	0.0000
joint.beta_occ_2_sf	-0.0104	0.1126	-0.09	0.9265	-10.0000	10.0000	0.0000
joint.beta_occ_3_sf	-0.5606	0.1290	-4.35	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_sf	0.7987	0.0920	8.68	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_2_cm	-1.4788	0.1135	-13.03	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_3_cm	-2.2247	0.1476	-15.07	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_cm	0.4714	0.0686	6.88	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_2_cf	0.1800	0.1000	1.80	0.0718	-10.0000	10.0000	0.0000
joint.beta_occ_3_cf	-0.2096	0.1113	-1.88	0.0597	-10.0000	10.0000	0.0000

